

Assessment for Delta Operations on Salmonids

Last updated: *January 28, 2026*

For more detailed data on salmonid conditions in the Delta see corresponding webpage on [SacPAS](#).

Executive Summary

- Entrainment management season is **active**.
- Season Loss: **0** (0.00% of threshold) DNA Winter-run, **0** (0.00% of threshold) Hatchery Winter-run, **47** (0.88% of threshold) Natural Steelhead, **280** (4.73% of threshold) Hatchery Steelhead, and **1019** (46.34% of threshold) Spring-run Surrogates.
- Winter-run presence in the Delta is **decreasing** (winding down).
- Steelhead presence in the Delta is **high** (historical peak).

Natural Winter-run Chinook

Juvenile Production Estimate

The Juvenile Production Estimate for winter-run is 1,057,452 for the current water year.

Current Status

Delta Entry Timing - Historically, as of Jan 27, 83% of length-at-date (LAD) winter-run have entered the Delta based on Knights Landing RST catch, 3% have exited the Delta based on Chipps Island Trawl Catch, and 2% of DNA confirmed winter-run have been salvaged.

Table 1: Average percent of annual emigrating population for unclipped LAD winter-run captured at monitoring locations and salvaged at Delta facilities for the past 10 years

Species	Red Bluff Diversion Dam	Tisdale RST	Knights Landing RST	Sac Trawl (Sher- wood)	Chipps Island Trawl	Salvage
Chinook, LAD Winter- run, Unclipped	99%	87%	83%	44%	3%	20%

Species	Red Bluff Diversion Dam	Tisdale RST	Knights Landing RST	Sac Trawl (Sher- wood)	Chipp Island Trawl	Salvage
Chinook, DNA Winter- run, Unclipped (Water Year)						2%

Red Bluff Diversion Dam Passage Estimate - As of Jan 21 estimated passage to date of LAD winter-run at Red Bluff Diversion is approximately 4.17 million fish. *Note that outmigration timing overlaps with spring-run migrating fish, and true winter-run abundance likely differs from these estimates.*

Delta Monitoring - Total catch of LAD winter run at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Jan 14 and Jan 25 is 41 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Jan 14 and Jan 26 is 2 individuals. Total catch at Delta Exit at Chippis Island between Jan 20 and Jan 27 is 0 individuals.

Annual Loss

The annual Loss threshold for natural winter-run is 1% of the jpe or 10,575 fish. As of January 27, cumulative loss of genetically confirmed winter-run is 0 or 0.00% of the annual loss threshold. Cumulative loss in the past 7 days has been 0.

STARS

The Delta STARS Model is an individual-based simulation model that predicts survival, travel time, and routing of juvenile salmon migrating through the Sacramento–San Joaquin River Delta. This model gives insight into survival and routing patterns of winter-run based on most current conditions.

As of January 27, overall through delta STARS estimated survival probability (with 80% credible intervals) is 0.53 (0.45-0.6) placing it in the 67th percentile of historical STARS survival estimates for the month of January (WYs 2018-2025). STARS estimated routing and survival probabilities (with 80% credible intervals) into the interior delta are 0.12 (0.1-0.14) and 0.23 (0.1-0.38), respectively, corresponding to the 33rd and 65th percentiles of historical January estimates (WYs 2018-2025).

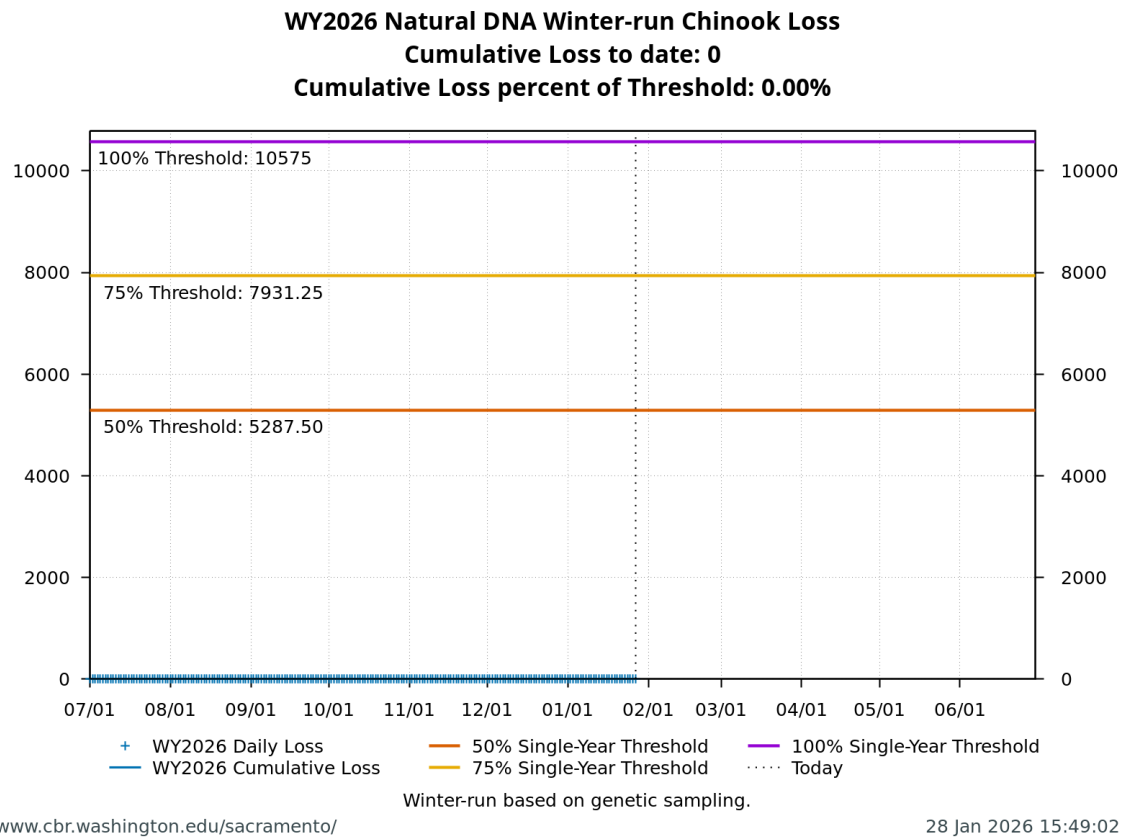


Figure 1: Cumulative loss of natural-origin winter-run for WY 2026. Cumulative loss is based on genetically confirmed winter-run captured in salvage or length-at-date winter-run in which genetic confirmation was unable to be obtained

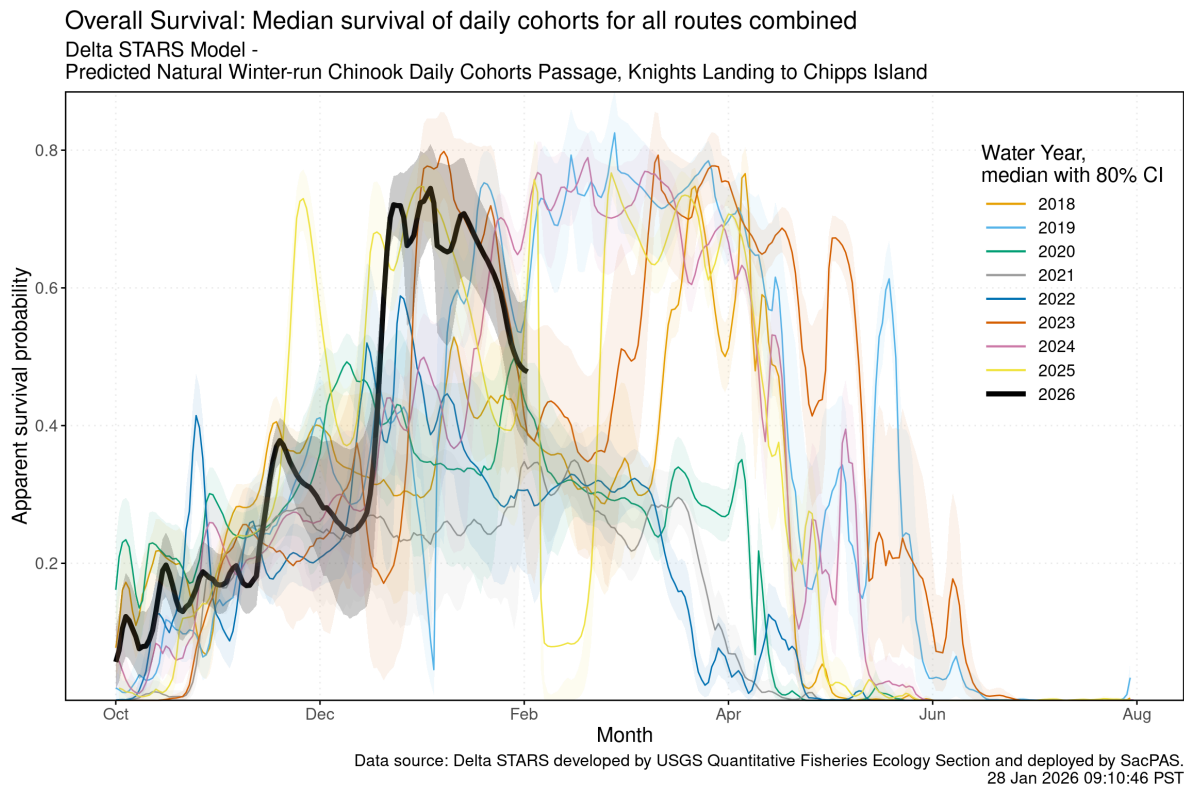


Figure 2: Estimated overall winter-run survival from Knights Landing to Chipps Island. Black line indicates the current water-year, and other colored lines correspond to past water years

Interior Delta Route-specific Survival Probability: Median survival of daily cohorts using the Interior Delta STARS Model - Predicted Natural Winter-run Chinook Daily Cohorts Passage, Knights Landing to Chipps Island

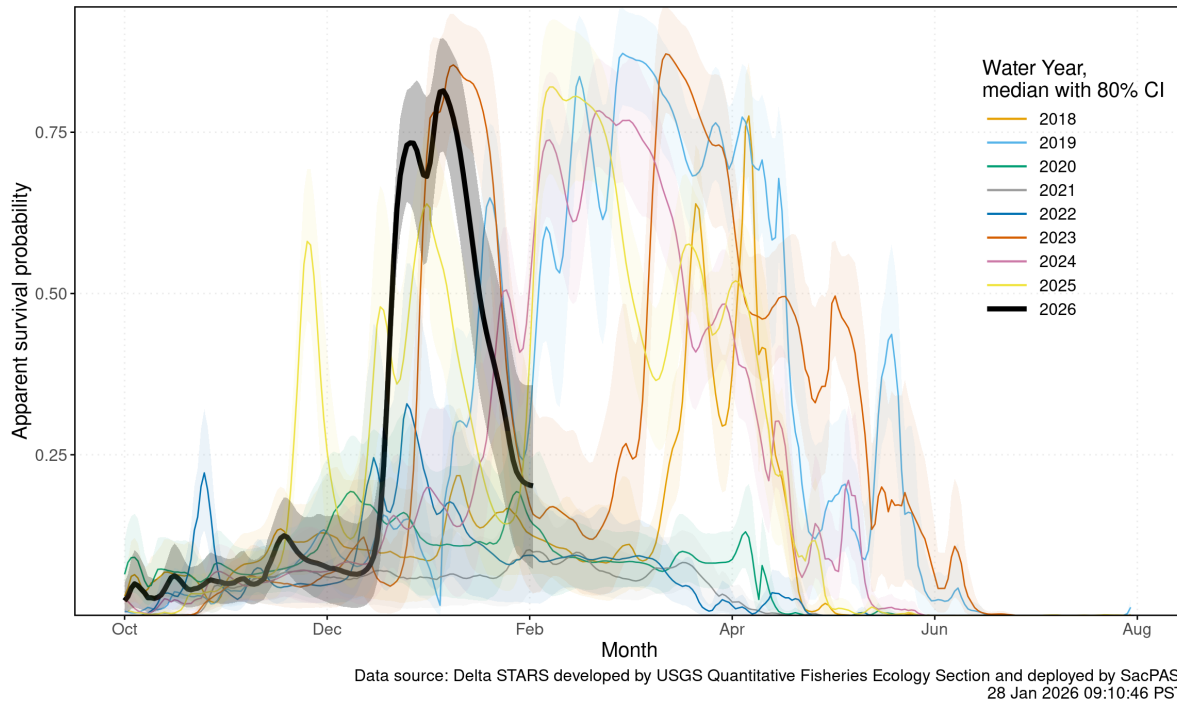


Figure 3: Estimated survival from Knights Landing to Chipps Island of simulated winter-run cohorts that route through the interior Delta. Black line indicates the current water-year, and other colored lines correspond to past water years

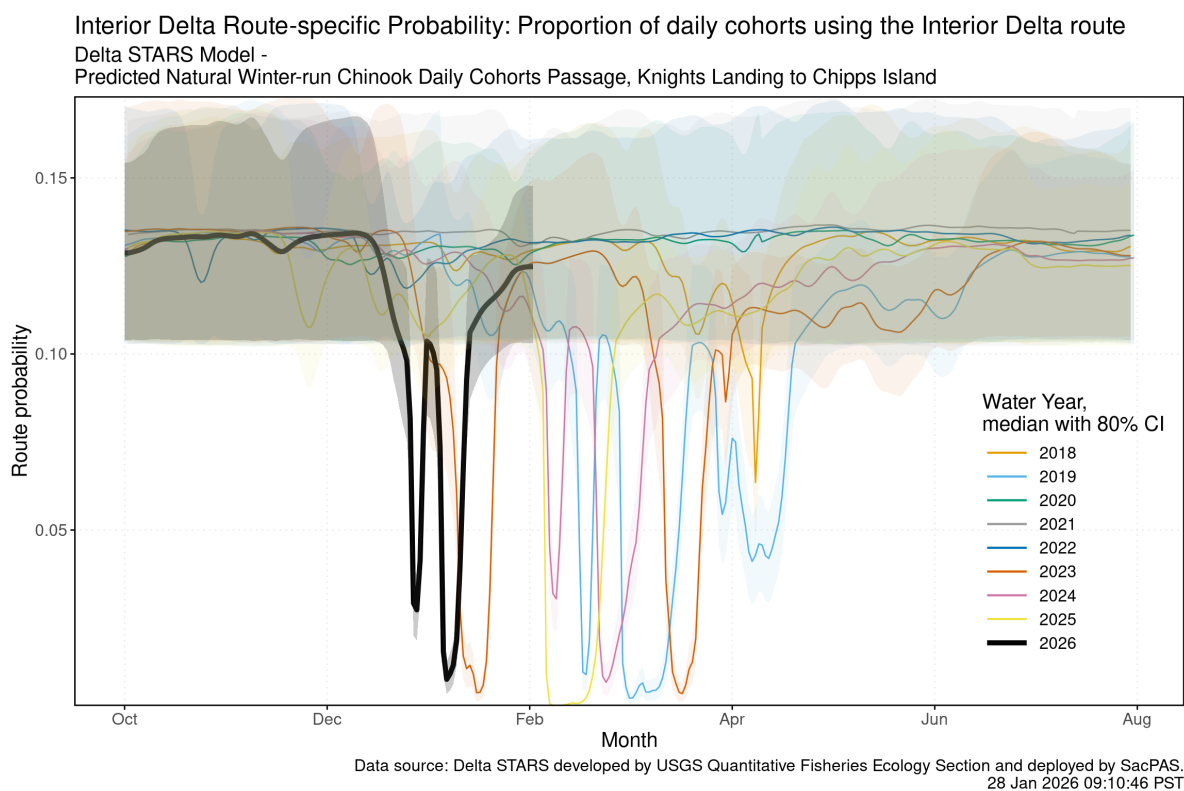


Figure 4: Estimated probability of winter-run routing into the interior Delta. Black line indicates the current water-year, and other colored lines correspond to past water years

Hatchery Winter-run Chinook

Hatchery Releases

To date, no winter-run Livingstone hatchery releases have occurred in WY 2026

Juvenile Production Estimate

The Juvenile Production Estimate for hatchery winter-run is 130,096 based on current Livingston Stone production estimates. **Note: Physical releases have not yet occurred in WY 2026.**

Annual Loss

To date, no loss has occurred as no hatchery winter-run have been released.

Natural-origin Central Valley Steelhead

Current Status

Delta Entry Timing - Historically, as of Jan 27, 32% of CCV steelhead have entered the Delta based on Knights Landing RST catch, 4% have exited the Delta based on Chipps Island Trawl Catch, and 9% have been salvaged.

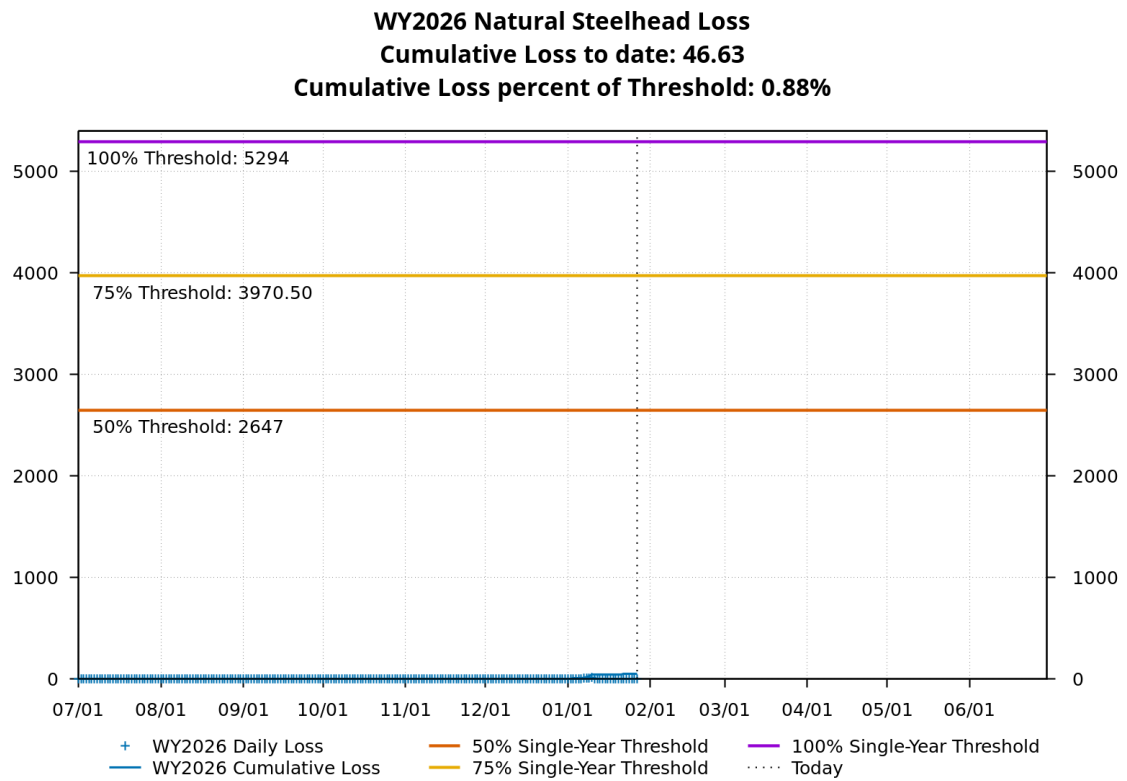
Table 2: Average percent of annual emigrating population for unclipped CCV steelhead captured at monitoring locations and salvaged at Delta facilities for the past 10 years

Species	Red Bluff Diversion Dam	Tisdale RST	Knights Landing RST	Sac Trawl (Sher- wood)	Chipps Island Trawl	Salvage
Steelhead, Unclipped	1%	24%	32%	6%	4%	9%

Delta Monitoring - Total catch of LAD winter run at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Jan 14 and Jan 25 is 3 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Jan 14 and Jan 26 is 0 individuals. Total catch at Delta Exit at Chipps Island between Jan 20 and Jan 27 is 0 individuals.

Annual Loss

As of January 27, cumulative loss of unclipped steelhead is 47 or 0.88% of the annual loss threshold. Cumulative loss in the past 7 days has been 127.



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Figure 5: Cumulative loss of natural-origin steelhead for WY 2026

Hatchery-origin Central Valley Steelhead

Surrogate Releases

There have been a total of 5 releases totaling 1,373,848 steelhead in Water Year 2026. JPE for the hatchery releases as of today is 591,419 based on estimated survivals using forecasted water year types (see details in table below). The annual loss threshold, equal to 1% of the JPE, is currently 5,914, but is subject to change with additional steelhead releases.

Table 3: Summary of steelhead hatchery releases in Water Year 2026. JPE calculated using hatchery-specific survival estimates to Delta entry from release location

Hatchery	Date of Release	Number Released	Estimated Survival	Juvenile Production Estimate
NIM	2025-11-10	233,109	72%	167,838
Coleman	2025-12-15	555,720	38%	211,174
Coleman	2025-12-17	90,019	38%	34,207
FRH	2026-01-06	371,250	36%	133,650
FRH	2026-01-09	123,750	36%	44,550

Table 4: Hatchery-specific survival estimates used for JPE calculations

Hatchery	Survival Estimate	Source
Coleman NFH	0.205 – 0.433	Sandstrom et al. 2020
Feather River Hatchery	0.09 – 0.45	Kurth 2013
Nimbus Hatchery	0.62 – 0.83	Brodsky et al. 2020

Hatchery	Survival Estimate	Source
Mokelumne River Hatchery	0.25 – 0.33	Del Real et al. 2012

Total loss of hatchery-origin steelhead is 280 or 4.73% of the annual loss threshold. *Note that hatchery origin of salvaged fish can not be determined at this time and salvage is based on the assumption of similar routing and survival probabilities of individual hatchery releases.*

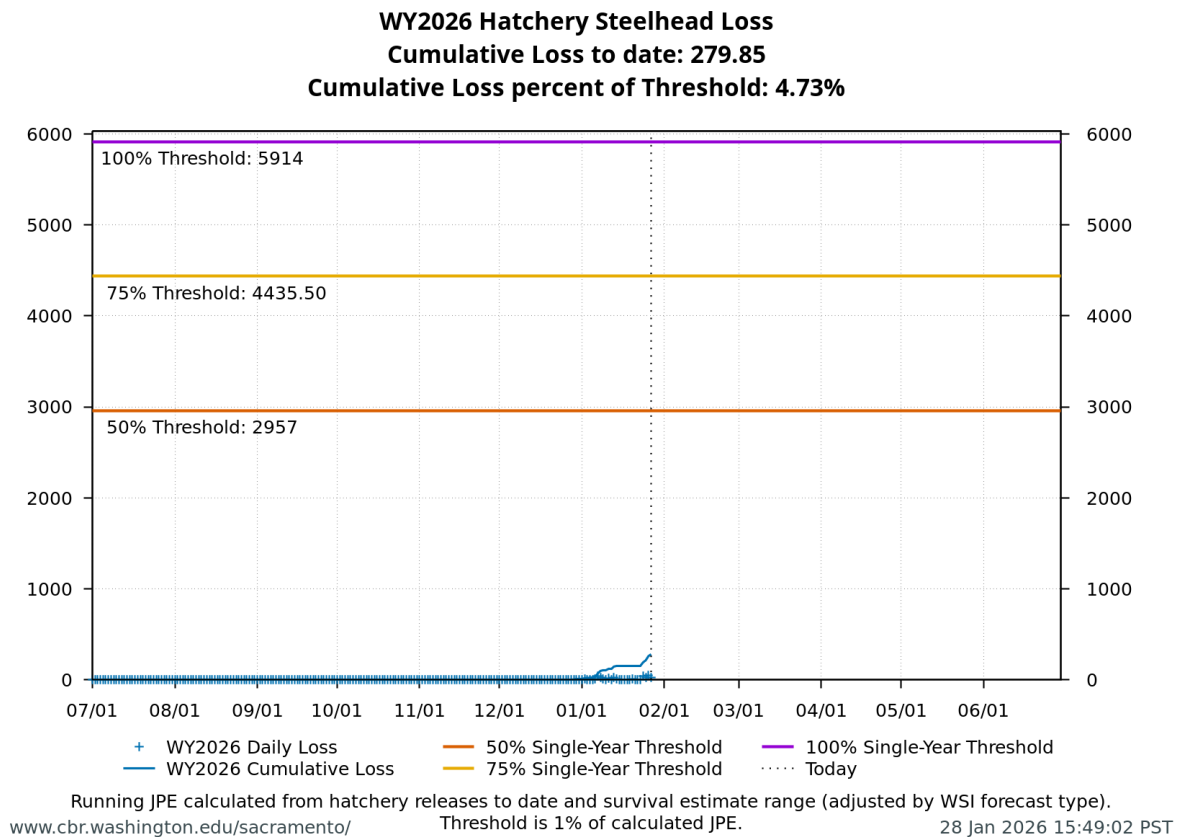


Figure 6: Cumulative loss of hatchery steelhead for WY 2026

Spring-run Chinook

Current Status

Delta Entry Timing - Historically, as of Jan 27, 28% of LAD spring-run have entered the Delta based on Knights Landing RST catch, 0% have exited the Delta based on Chipps Island Trawl Catch, and 0% have been salvaged.

Table 5: Average percent of annual emigrating population for LAD spring-run Chinook salmon captured at monitoring locations and salvaged at Delta facilities for the past 10 years

Species	Red Bluff Diversion Dam	Tisdale RST	Knights Landing RST	Sac Trawl (Sher- wood)	Chipps Island Trawl	Salvage
Chinook, LAD Spring- run, Unclipped	14%	16%	28%	2%	0%	0%

Red Bluff Diversion Dam Passage Estimate - As of Jan 21 estimated passage to date of LAD spring-run at Red Bluff Diversion is approximately 0.02 million fish. *Note that outmigration timing overlaps with winter-run and fall-run outmigration, and true spring-run abundance likely differs from these estimates.*

Delta Monitoring - Total catch of LAD winter run at RSTs at Delta Entry (Tisdale, Knights Landing, Lower Sacramento River) between Jan 14 and Jan 25 is 164 individuals. Total catch at Sacramento Trawl and Beach Seines in the delta between Jan 14 and Jan 26 is 0 individuals. Total catch at Delta Exit at Chipps Island between Jan 20 and Jan 27 is 0 individuals.

Spring-run Surrogate Releases

A total of 805,323 spring-run surrogate fish have been released in Water Year 2026, with an estimated Juvenile Production Estimate (JPE) of 219,852 fish entering the Delta. This includes 805,323 Coleman Late-Fall Run Chinook (JPE: 219,852) released from Coleman National Fish Hatchery across 12 coded-wire tag groups. See details in table below.

Table 6: Spring-run Chinook salmon surrogate releases (all Coleman Late-Fall releases, both production and experimental)

Hatchery	Release Date	Type	# of CWT Fish Released	JPE	Confirmed CWT Loss Codes
Coleman NFH	2025-11-13	Production	143,346	39,134	8.7 056808, 056809
Coleman NFH	2025-11-17	Experimental	75,119	20,507	0.0 056810
					053700, 056806, 056811, 056812, 056814, 056815, 056817
Coleman NFH	2025-12-17	Production	468,876	128,002	749.9 056812, 056814, 056815, 056817
Coleman NFH	2025-12-22	Experimental	60,873	16,618	257.3 056813
Coleman NFH	2026-01-08	Experimental	57,109	15,591	2.9 056816

Annual Loss

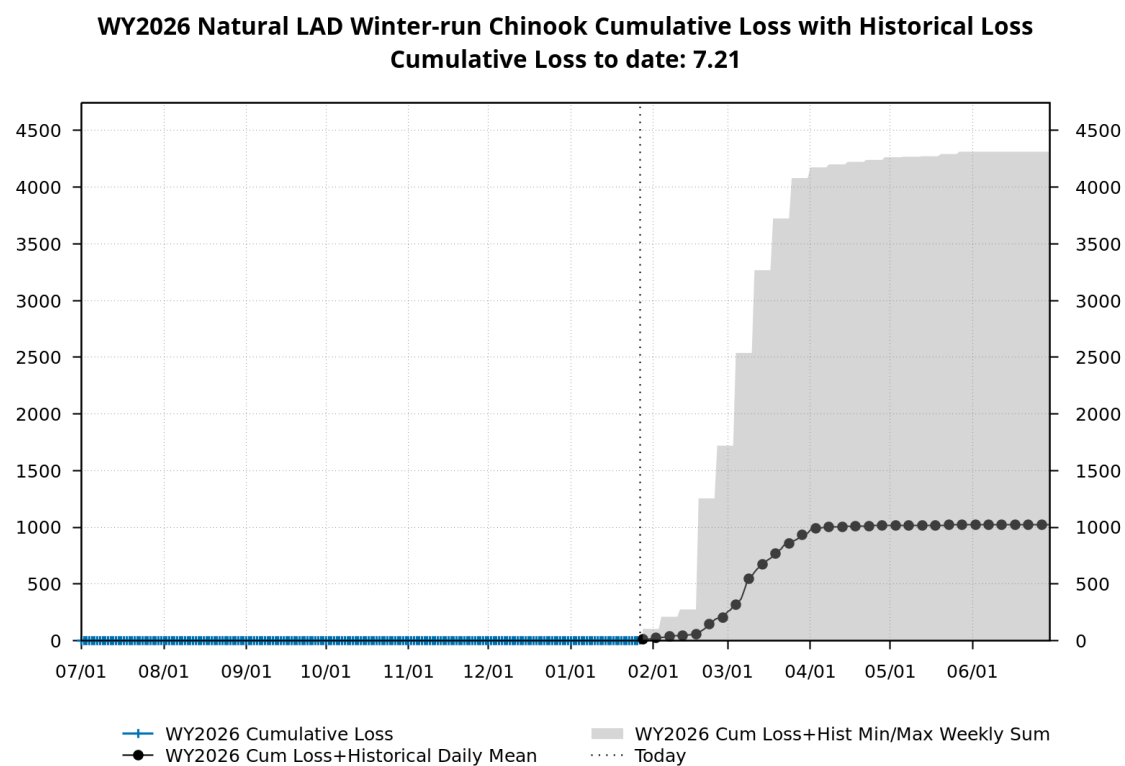
The annual loss threshold is 1% of the JPE entering the Delta, which equals 2,199 fish. As of January 27, cumulative loss is 1,019 fish or 46.34% of the annual loss threshold.

Loss Prediction and Trajectories

The following figures display the cumulative loss trajectories relative to historical years (“Spaghetti Plots”) and the current loss predictor model outputs for winter-run Chinook salmon and steelhead.

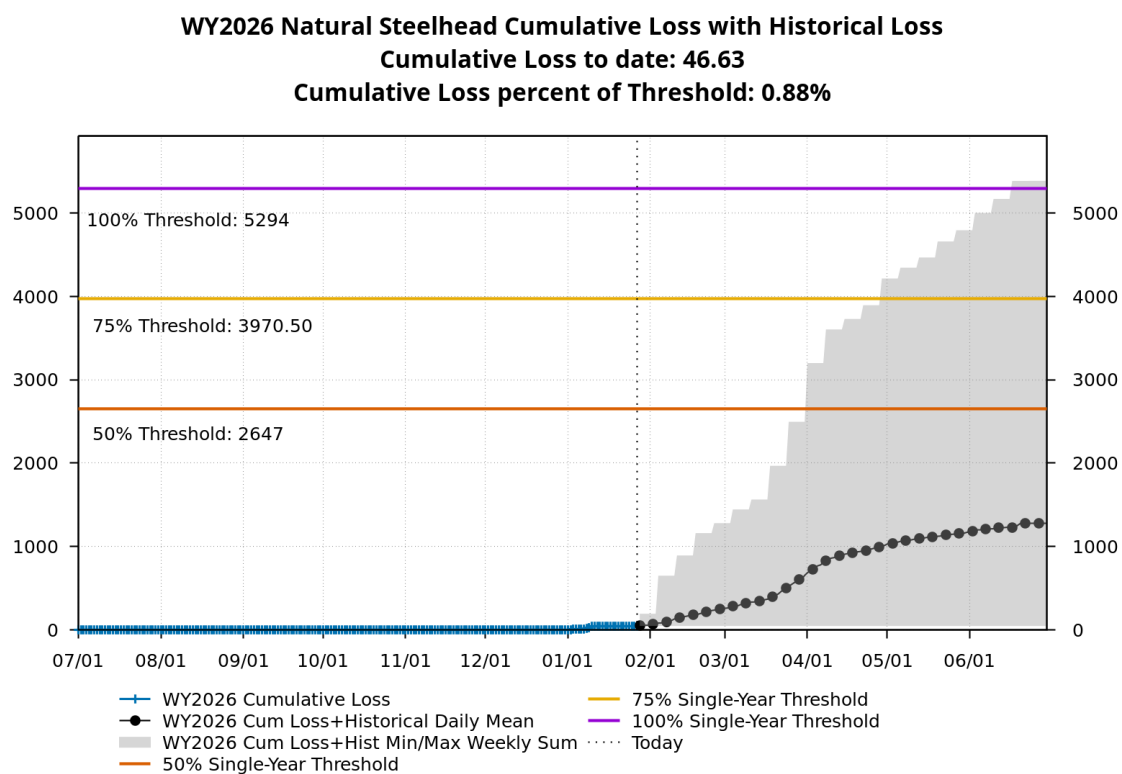
Evaluation

1. What is the probability of exceeding natural or hatchery winter-run Chinook salmon loss thresholds in the upcoming week?



Winter-run based on Length-at-Date Delta Model. Historical Loss calculated for WY2009-2018.
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Figure 7: Natural LAD winter-run Chinook salmon loss cumulative to date with historical years and annual loss threshold



Historical Loss calculated for WY2009-2018.

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Figure 8: Natural Central Valley steelhead loss cumulative to date with historical years and annual loss threshold

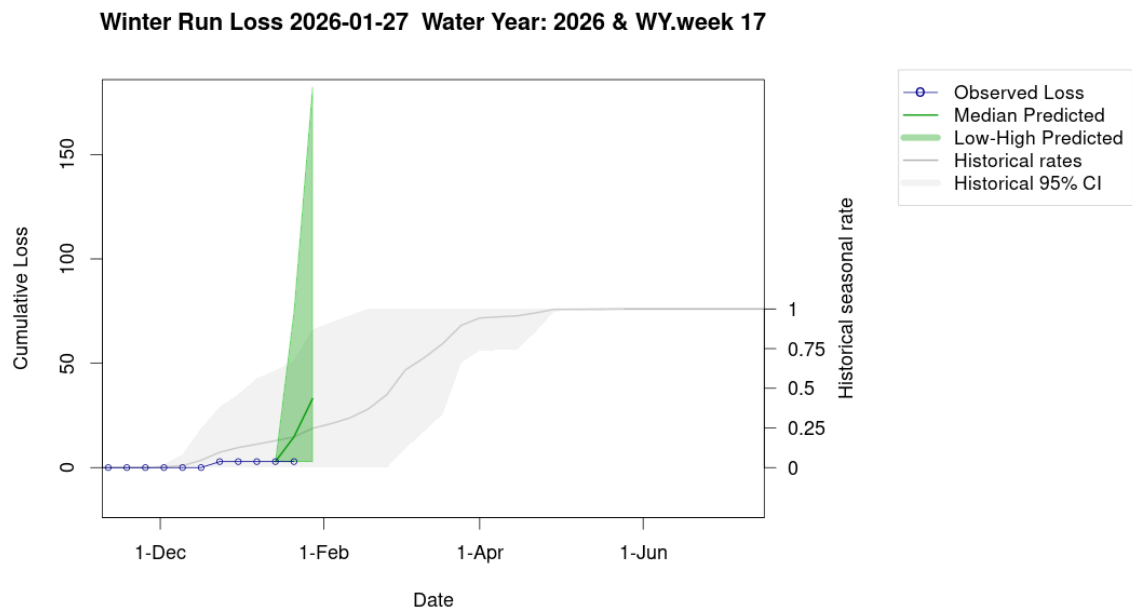


Figure 9: Estimates of winter-run Chinook loss generated by Loss and Salvage Predictor tool

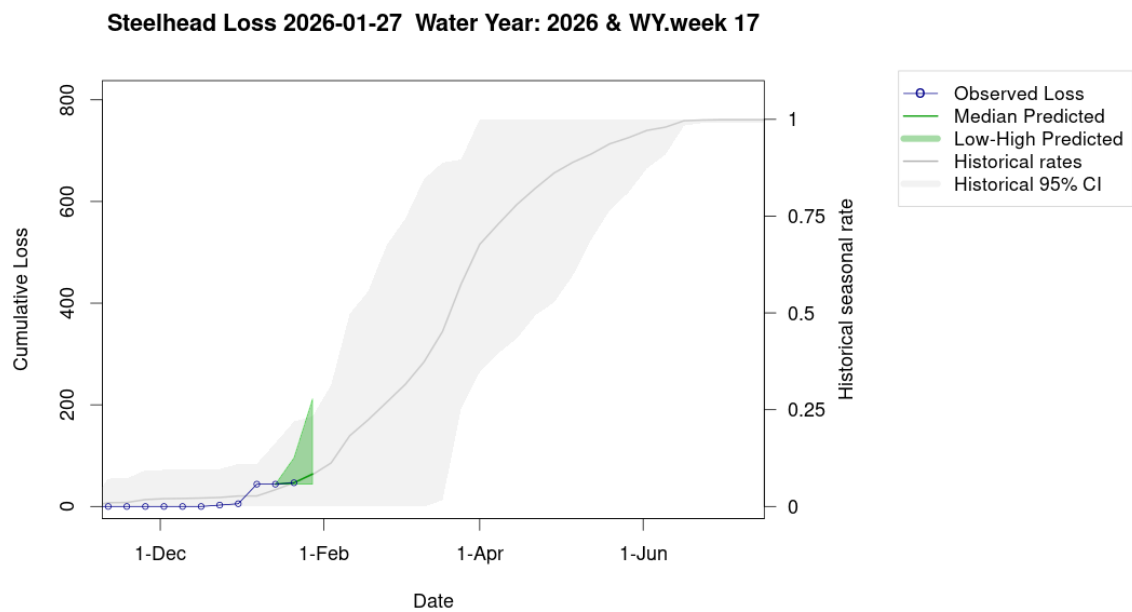


Figure 10: Estimates of steelhead loss generated by Loss and Salvage Predictor tool

LOW RISK: Cumulative loss is currently 0% of the threshold. Current trajectory suggests the threshold is unlikely to be exceeded in the upcoming week.

2. **What is the probability of spring-run hatchery Chinook salmon loss thresholds in the upcoming week?**

LOW RISK: Cumulative loss is currently 46.3% of the threshold. Current trajectory suggests the threshold is unlikely to be exceeded in the upcoming week.

3. **What is the probability of hatchery steelhead loss thresholds in the upcoming week?**

LOW RISK: Cumulative loss is currently 4.7% of the threshold. Current trajectory suggests the threshold is unlikely to be exceeded in the upcoming week.

References

- Brodsky, A., Zeug, S. C., Nelson, J., Hannon, J., Anders, P. J., & Cavallo, B. J. (2020). Does broodstock source affect post-release survival of steelhead? Implications of replacing a non-native hatchery stock for recovery. *Environmental Biology of Fishes*, 103(5), 437–453.
- Del Real, S. C., Workman, M., & Merz, J. (2012). Migration characteristics of hatchery and natural-origin oncorhynchus mykiss from the lower mokelumne river, california. *Environmental Biology of Fishes*, 94(2), 363–375.
- Kurth, R. (2013). *Downstream migration success of feather river fish hatchery steelhead smolts under different release strategies*.
- Sandstrom, P. T., Ammann, A. J., Michel, C., et al. (2020). Low river survival of juvenile steelhead in the sacramento river watershed. *Environmental Biology of Fishes*, 103(5), 531–541.